

Title: Detection of Scam Transactions Involving Mule Accounts in Thai Banks

1. Problem Statement/Background Online scams have surged dramatically. In 2024, total damages exceeded 60,000–80,000 million THB (data from Cyber Police / Whoscall) • Mule accounts are increasing significantly, especially among teenagers and economically disadvantaged groups • Victims are often persuaded to voluntarily transfer money themselves → very difficult to detect • Current systems are slow — money is already gone before intervention occurs • Banks are under pressure from the Bank of Thailand (BOT), media, and society	2. SMART Objectives/ Value Propositions SMART Objective: To identify and detect mule account transactions by analyzing transaction patterns, customer profiles, and risk factors, in order to increase the detection rate of suspicious transactions by at least 20% within 12 months	3. Questions/Hypothesis Key Questions: 1. Which customers are most likely to be mule accounts? 2. What transaction patterns indicate mule behavior ? 3. When do suspicious transactions occur most frequently ? 4.Which countries are most associated with suspicious transactions? 5.Why do customers with low MonthlyIncome conduct high-value transactions? 6.How can transaction patterns be used to identify mule accounts? Hypotheses: H1. Customers with lower MonthlyIncome are more likely to be involved in suspicious transactions. H2. Accounts with many incoming transactions followed by immediate outgoing transactions are likely to be mule accounts. H3. Suspicious transactions occur more frequently during weekends (IsWeekend = 1). H4. Transactions involving countries with high AMLScore are more likely to be suspicious. H5. Customers with low MonthlyIncome but high transaction amounts are likely acting as mule accounts. H6. Accounts with a high number of unique senders are more likely to be mule accounts.	4. Key Metrics/Attributes H1 Metrics: % IsSuspicious by income, Avg MonthlyIncome Attributes: =MonthlyIncome, IsSuspicious H2 Metrics: Time gap (in/out), #transactions Attributes: SenderAccountKey, ReceiverAccountKey, Date H3 Metrics: % IsSuspicious (Weekend vs Weekday) Attributes: IsWeekend, DayOfWeek H4 Metrics: % IsSuspicious by AMLScore Attributes: AMLScore, CountryKey H5 Metrics: Amount / MonthlyIncome, % Amount > Income Attributes: Amount, MonthlyIncome H6 Metrics: #Unique senders, % IsSuspicious Attributes: SenderAccountKey, ReceiverAccountKey
5. Analysis/Model Prepare 1. Dataset Creation → Generated a cleansed and realistic dataset , Designed to support meaningful analysis 2. Data Volume → Fact table (Transaction Logs): → Maximum 1,000 rows 3. Fraud Scenario Simulation → Simulated Money Mule accounts , Represent ~1–3% of total data 4. Clean data → remove null , check data type and make it be accurate Analytical actions • Exploratory Data Analysis (EDA) → Perform data patterns to find the pattern of data like behavior of mule or no mule • Segment data → dividing a dataset into smaller groups based on shared traits like income, behavior or location. • Test hypotheses using metrics → using quantifiable data points (KPIs) to "validate" or "refute" assumptions. • Identify anomalies and mule account patterns → detecting data points that deviate from standard customer behavior (Anomalies) and recognizing specific signatures of "Money Mules" (Patterns). Key indicators include "Rapid Flow-through" • Elimination and Prevention → A dual-strategy approach to breaking the money mule cycle. Elimination focuses on detecting and neutralizing active mule accounts by identifying behavioral triggers	6. Findings and Insights Key Questions 1.Low-income individuals are more likely to become mule account holders, as they may not have sufficient financial resources and are therefore more willing to sell or lend their bank accounts. 2.Mule accounts tend to transfer out large amounts of money while keeping only a small remaining balance in the account, in case the account gets detected and closed. 3.June tends to have a higher number of mule accounts and higher transaction liquidity, due to school fee payments and mid-year financial activities, which create opportunities for mule accounts to blend in more easily. 4.Despite a low AML Score (perceived as safe), Thailand shows the highest volume of mule activity. This indicates a "High-Volume, Low-Score" anomaly, proving that official country risk ratings do not capture the actual operational risk. 5.The Mule Score results reveal that most mule accounts are not classified as "High Risk" but fall into the "Potential Risk" category. This suggests that mules are increasingly sophisticated, mimicking legitimate behavior to stay below the maximum danger threshold.	7. Recommendation/Action and Impact 1. Behavioral-Based Detection → Recommendation: Shift focus from static income data to Behavioral Analytics, specifically monitoring the "In-Out Velocity" and low ending balances. Impact: Increases detection accuracy of "Sophisticated Mules" who stay below traditional risk thresholds but display high-frequency movement. 2.Seasonal & Regional Calibration → Recommendation: Implement Dynamic Thresholds for high-activity periods (June) and override global AML scores with a Localized Risk Rating for Thailand. Impact: Closes the gap between official risk ratings and operational reality, reducing the "High-Volume, Low-Score" anomaly. 3.Predictive Outreach & Defense → Recommendation: Launch targeted Awareness Campaigns for low-income segments and deploy real-time alerts during transaction peaks. Impact: Prevents vulnerable individuals from being recruited as mules and minimizes financial loss through proactive intervention.	